UNIVERSITY FINAL PROJECT • MACHINE LEARNING

Fake Job Postings Detector

A Machine Learning Approach to Screen Online Job Ads



Presenter Name · University Name
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A trust problem hiding in plain sight

Fake ads exploit urgency — and trust.

They waste applicants' time, expose personal information, and scale quickly across global job platforms.

The need

Automated screening that helps people and platforms review risk earlier.



ONLINE JOB AD → **ML RISK SCREEN** → **REAL / FLAGGED**

Classify each posting

Combine text + structured signals

Escalate suspicious ads for review

A triage tool — not a final decision-maker

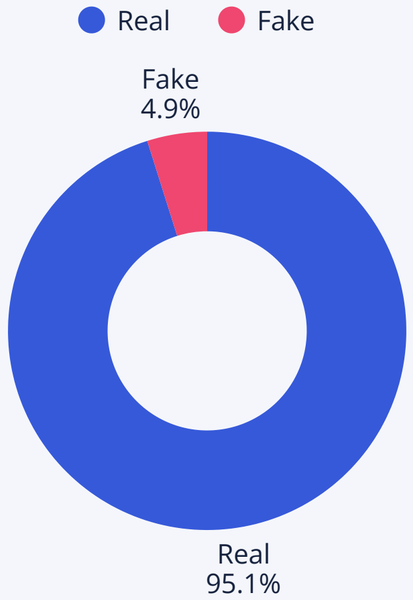
17,599
unique postings

856
fake postings

18
input columns

TARGET ‘fraudulent’ • 1 = fake • 0 = real

Class distribution



Only 4.86% are fake

Two views of every job ad

TEXT SIGNALS

Title
Description
Requirements
Benefits
Company profile

STRUCTURED SIGNALS

Logo present
Screening questions
Contact information
Location
Employment type

Four signals stood out

01

Logos

Fake ads are less likely to show a company logo.

02

Screening

Screening questions appear less often in fake ads.

03

Contact

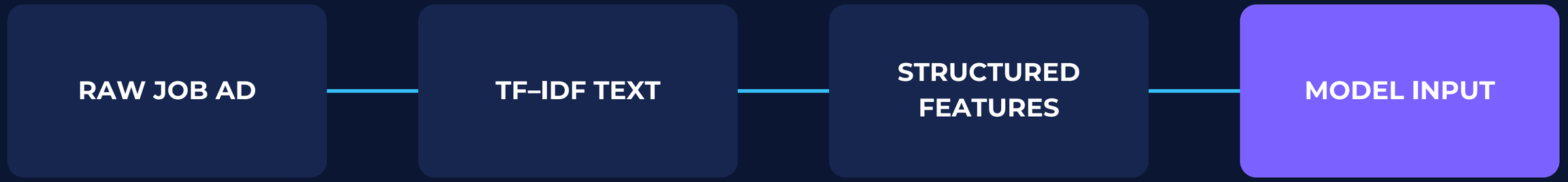
Email or phone details appear more often in fake ads.

04

Repetition

Duplicates and near-duplicates reveal recurring campaigns.

From raw advert to model-ready features



Words become weighted numeric signals

Logo • profile • contact • length • country

Three models. One shared objective.

01

LOGISTIC REGRESSION

Strong interpretable baseline

02

LINEAR SVM

Excellent for sparse text

03

RANDOM FOREST

Non-linear structured patterns

Class weights counter the rarity of fake postings

Evaluate for the failure that matters

GROUP-AWARE TRAIN SPLIT

HELD-OUT TEST SPLIT

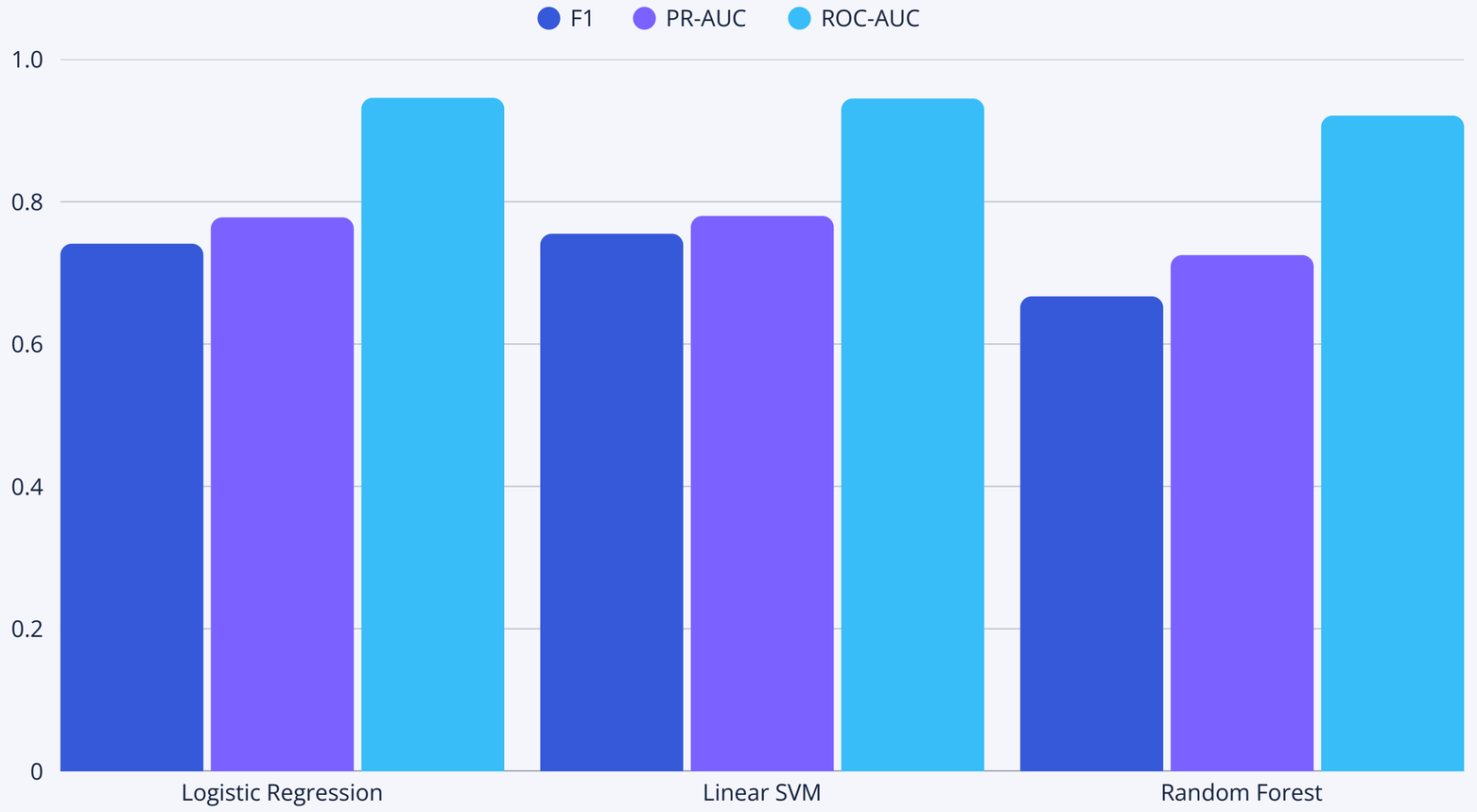
Near-duplicate ads stay together → less leakage

PR–AUC

Best summary when fake ads are rare

PRECISION RECALL F1 ROC–AUC

Linear models lead the field



BEST F1

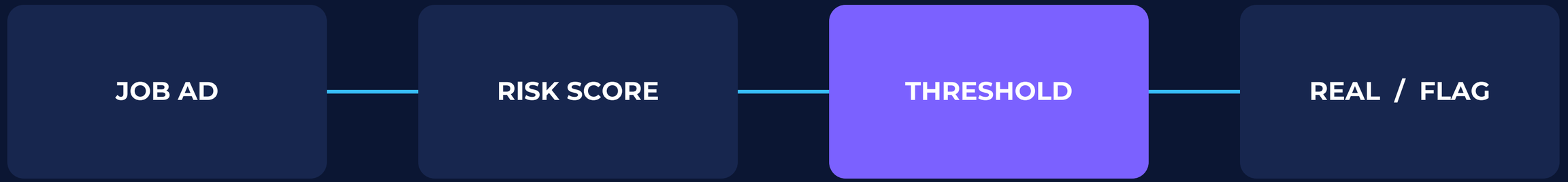
0.755

Linear SVM

95.1% accuracy baseline

Accuracy is misleading under severe class imbalance.

A score becomes an actionable flag



The cutoff is tuned with out-of-fold training predictions — never on the final test set.

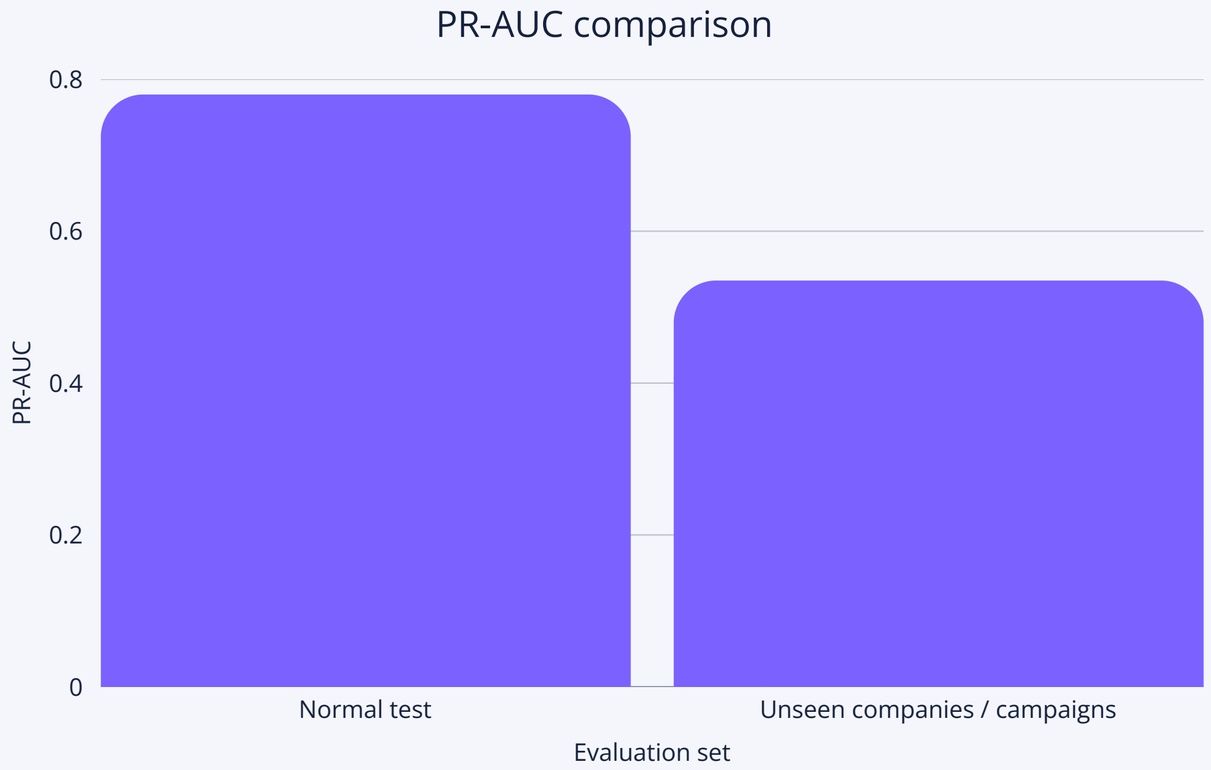
The model combines weak signals

- **Missing company profile**
- **No logo**
- **No screening questions**
- **Direct contact details**
- **Recurring scam language**

No single ‘magic’ word

Predictions emerge from patterns across content and metadata. Some patterns still reflect repeated templates or dataset-specific terms.

Novel scams expose the generalization gap



0.53–0.54

PR–AUC on unseen scams

Strong on known patterns. Weaker on completely new campaigns.

Use as triage — not certainty.

FINAL APPLICATION

Safer screening, clearer review.

A user-friendly web app turns the model into an accessible triage tool for job seekers and platforms.

NEXT • Monitor drift • Refresh scam patterns • Keep humans in the loop



Thank you — Questions?